

Lab 15: Inferring the Secret Ballot

84-355 Democracy's Data | Racially polarized voting, Houston County, Georgia

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Everything here runs as given. Your job is to read what comes out and say what it means.

Before we start

Section 2 of the Voting Rights Act turns on a factual question: **do voters of different races prefer different candidates?** If they do not, there is no claim. Everything you read this morning — King, Monmonier — exists because of one obstacle in the way of answering it.

Note — the law under this changed on 29 April 2026

Louisiana v. Callais, 608 U.S. ____ (2026), did not overrule *Gingles*. It updated it, and the update lands on the two preconditions this lab is about. A plaintiff must now “provide an analysis that controls for party affiliation” and show racial bloc voting “that cannot be explained by partisan affiliation.” Simply pointing to inter-party polarization, the Court said, “proves nothing.” Five weeks later it put it flatly: “the mere fact that voters of different races vote for different parties is not relevant to proving racially polarized voting patterns,” *Allen v. Milligan*, 608 U.S. ____ (2026) (per curiam) (June 2, 2026).

What we estimate today is which party's ballot each racial group asked for — the inter-party comparison, exactly. The statistical problem is unchanged and everything below still runs. What changed is what the answer would be worth in court, and that is worth holding in mind from Part 6 on.

The ballot is secret. Nobody records how you voted. What gets recorded is how a *precinct* voted, and separately, who *lives* in that precinct. From those two aggregates an expert witness must infer how each racial group behaved — and then a federal judge decides a case on the answer.

That inference is called **ecological inference**, and it is the thing every statistics course warns you about: reasoning from groups to individuals. Do it badly and you commit the ecological fallacy.

Today we do it, and then we find out whether it worked.

Why we can check. In Georgia, two facts are public that are private almost everywhere else: the race a voter reports on the registration form, and **which party's primary ballot they requested**. So for one election — the General Primary of 21 May 2024 — we know, person by person, the exact quantity ecological inference is trying to estimate.

This is the same trick as last week. Week 14 graded a guess about *who someone is*. Today we grade a guess about *what they did*.

For the next four parts, **pretend we do not know the answer**. That is the honest position of every expert who has ever testified on this.

Part 1 — What you observe

```
p <- read.csv("data/houston_primary.csv", stringsAsFactors = FALSE)

p$pct_black <- p$black / p$voters
p$pct_dem   <- p$dem   / p$voters

obs <- p[order(p$pct_black), c("precinct", "voters", "pct_black", "pct_dem")]
print(obs, row.names = FALSE, digits = 3)
```

```
#>                precinct voters pct_black pct_dem
#>                HEFS      375    0.109    0.115
#>                HAFS      465    0.114    0.131
#>                ROZR     1606    0.140    0.163
#>                BMS      1388    0.192    0.233
#>                CGTC      880    0.225    0.257
#>                ANNX      825    0.236    0.276
#>                PEC      1092    0.240    0.273
#>                VHS      1296    0.252    0.270
#>                VFW       571    0.256    0.327
#>                MCMS     1413    0.299    0.306
#>                TWPK     1095    0.301    0.347
#> NORTH HOUSTON SPORT COMPLEX  424    0.314    0.325
#>                CENT     1008    0.344    0.372
#>                HHPC     1058    0.359    0.421
#>                FMMS     1248    0.369    0.382
#>                TMS       508    0.500    0.561
#>                WELLSTON CENTER  772    0.766    0.771
```

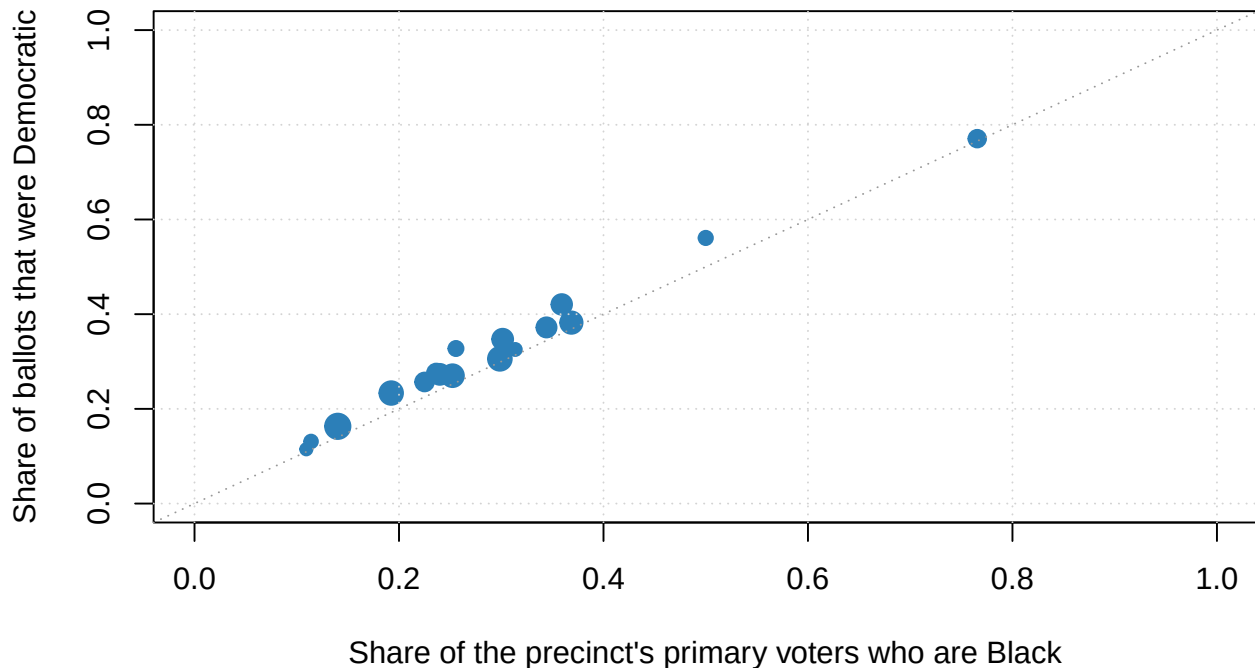
Seventeen precincts, 16,024 voters. Two columns matter: what share of each precinct's primary electorate was Black, and what share of its ballots were Democratic.

This table is the entire evidentiary record. An expert witness in a Section 2 case gets essentially this and nothing more.

Notice the range. The least Black precinct (HEFS) is 10.9% Black; the most (WELLSTON CENTER) is 76.6%. **There is no precinct in this county that is overwhelmingly Black.** Hold on to that — it matters more than it looks.

Part 2 — The eyeball

```
plot(p$pct_black, p$pct_dem, pch = 19, cex = sqrt(p$voters) / 22,
     col = "#2c7fb8", xlim = c(0, 1), ylim = c(0, 1),
     xlab = "Share of the precinct's primary voters who are Black",
     ylab = "Share of ballots that were Democratic")
abline(0, 1, lty = 3, col = "grey60")
grid()
```



The relationship is about as clean as social science ever gets. Precincts with more Black voters cast more Democratic ballots, and the points sit close to the 45-degree line — which would be exactly what you would see if **every** Black voter took a Democratic ballot and **no** white voter did.

That is a hypothesis, not a finding. Two different worlds produce this picture: one where Black and white voters are perfectly polarized, and one where Democratic-leaning people of every race happen to live near each other. The scatter cannot tell them apart. **Neither can any method in this lab.**

Part 3 — Homogeneous precincts

The oldest method in voting-rights litigation, and still the one courts find most intuitive: **find precincts that are nearly all one race, and just look at how they voted.** No modelling, no assumptions.

```
hb <- p[p$pct_black >= 0.80, ]
hw <- p[p$pct_black <= 0.20, ]

cat("precincts at least 80% Black:", nrow(hb), "\n")
```

```
#> precincts at least 80% Black: 0
```

```
cat("precincts at most 20% Black:", nrow(hw), "\n\n")
```

```
#> precincts at most 20% Black: 4
```

```
if (nrow(hw))
  cat(sprintf("white-homogeneous precincts, Democratic share: %.1f%%\n",
              100 * sum(hw$dem) / sum(hw$voters)))
```

```
#> white-homogeneous precincts, Democratic share: 18.0%
```

```
if (nrow(hb))
  cat(sprintf("Black-homogeneous precincts, Democratic share: %.1f%%\n",
             100 * sum(hb$dem) / sum(hb$voters)))
```

Read that first number again: there are none. Not one precinct in Houston County is 80% Black, so the most trusted method in the field cannot produce an estimate for Black voters here *at all*. Not an imprecise one — none.

This is not a quirk of our threshold. It is a fact about the county, and it is the ordinary situation in most of the United States. The method works best exactly where segregation is most severe.

And the number it *does* give us is worth distrusting. The 4 “white-homogeneous” precincts return **18.0%** Democratic — but those precincts are up to 20% Black, and those Black voters cast ballots too. The estimate for white voters is contaminated by the very group we are trying to hold out.

Part 4 — Goodman’s ecological regression

The standard alternative, and for fifty years the workhorse of Section 2 litigation. Fit a line through the scatter, then read off its ends: where the line crosses `pct_black = 0` is the estimate for white voters, and where it crosses `pct_black = 1` is the estimate for Black voters.

```
fit <- lm(pct_dem ~ pct_black, data = p, weights = voters)
b <- coef(fit)

est_white <- 100 * b[1]
est_black <- 100 * (b[1] + b[2])

cat(sprintf("R-squared: %.3f\n\n", summary(fit)$r.squared))
```

```
#> R-squared: 0.982
```

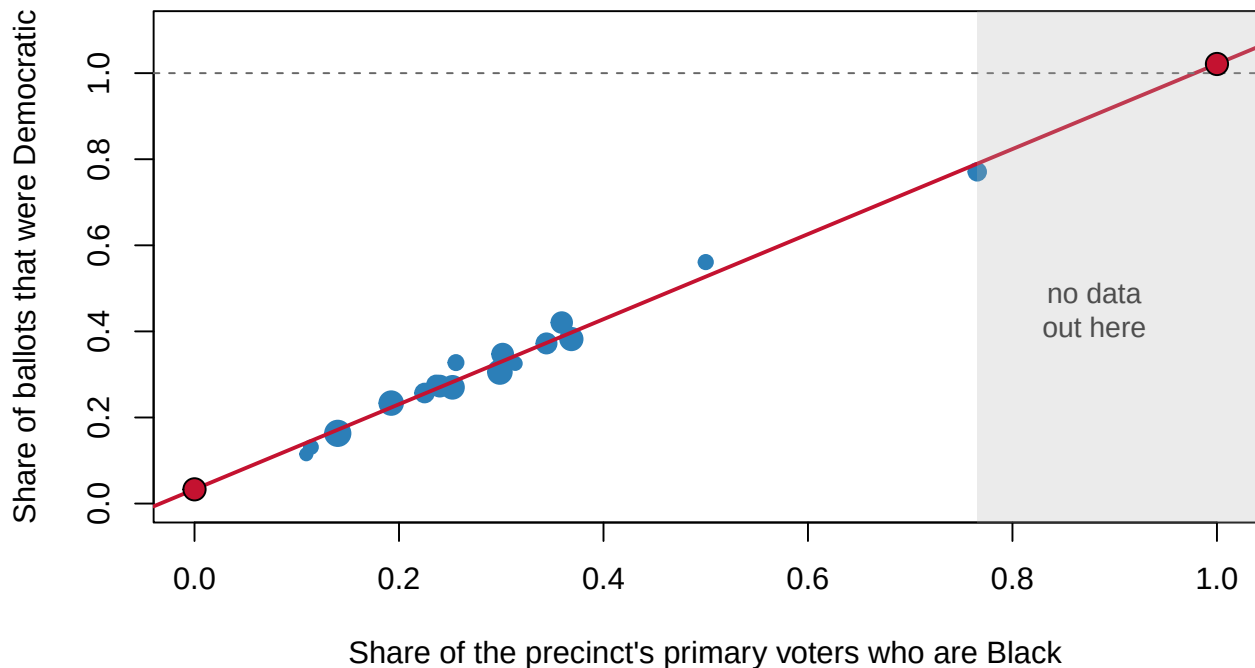
```
cat(sprintf("  estimated Democratic support, white voters: %5.1f%%\n", est_white))
```

```
#>   estimated Democratic support, white voters:   3.3%
```

```
cat(sprintf("  estimated Democratic support, Black voters: %5.1f%%\n", est_black))
```

```
#>   estimated Democratic support, Black voters: 102.1%
```

```
plot(p$pct_black, p$pct_dem, pch = 19, cex = sqrt(p$voters) / 22,
     col = "#2c7fb8", xlim = c(0, 1), ylim = c(0, 1.1),
     xlab = "Share of the precinct's primary voters who are Black",
     ylab = "Share of ballots that were Democratic")
abline(fit, col = "#C41230", lwd = 2)
abline(h = 1, lty = 2, col = "grey40")
rect(max(p$pct_black), -1, 1.2, 2, col = adjustcolor("grey70", alpha.f = 0.25),
     border = NA)
text(0.88, 0.45, "no data\nout here", col = "grey30", cex = 0.9)
points(c(0, 1), c(b[1], b[1] + b[2]), pch = 21, bg = "#C41230", cex = 1.6)
```



The model fits beautifully — R^2 of 0.982 — and returns an impossible answer. It estimates that 102.1% of Black voters took a Democratic ballot.

There is no gentle reading of this. A percentage above 100 is not a close call or a rounding problem; it is a signal that the method has been asked to do something it cannot do. And notice that nothing in the regression *complained*. The fit statistics are superb. A 0.98 R^2 is the kind of number that gets quoted in a report.

The grey band shows why. **The most Black precinct in the county is 76.6% Black**, so the estimate for Black voters comes from extending the line 23.4 points past the last observation. The model is not summarising data out there. There is no data out there.

Part 5 — What is knowable without a model

Before King, Duncan and Davis pointed out something simple: even with no model at all, the precinct totals *constrain* the answer. If a precinct is 40% Black and cast 45% Democratic ballots, then even if every single white voter went Democratic, Black voters must supply the rest — and that puts a floor and a ceiling on their behaviour.

```
bl <- p$pct_black
dm <- p$pct_dem

lo <- pmax(0, (dm - (1 - bl)) / bl) # if every white voter went Democratic
hi <- pmin(1, dm / bl)             # if no white voter did

cat(sprintf("Tightest single precinct: [%.1f%%, %.1f%%] (%s)\n",
            100 * max(lo), 100 * min(hi), p$precinct[which.min(hi - lo)]))
```

```
#> Tightest single precinct: [70.1%, 100.0%] (WELLSTON CENTER)
```

```
cat(sprintf("Intersection of all 17:  [%.1f%%, %.1f%%]\n",
          100 * max(lo), 100 * min(hi)))
```

```
#> Intersection of all 17:  [70.1%, 100.0%]
```

```
cat(sprintf("width: %.1f points\n", 100 * (min(hi) - max(lo))))
```

```
#> width: 29.9 points
```

Black Democratic support is between 70.1% and 100%. That statement requires no assumptions whatsoever — it is arithmetic on the marginals, and it cannot be wrong.

It is also **29.9 points wide**, which is why nobody testifies with it alone.

Here is the trade the whole field runs on. The bounds are *guaranteed correct and too vague to decide a case*. The regression is *precise, decisive, and impossible*. King's method — the reading for today — is an attempt to get something in between: use the bounds as a hard constraint, then use a statistical model to say where inside them the answer probably sits.

Part 6 — The answer

We know the truth. Every voter in that table requested a party ballot by name, and reported their race on the registration form.

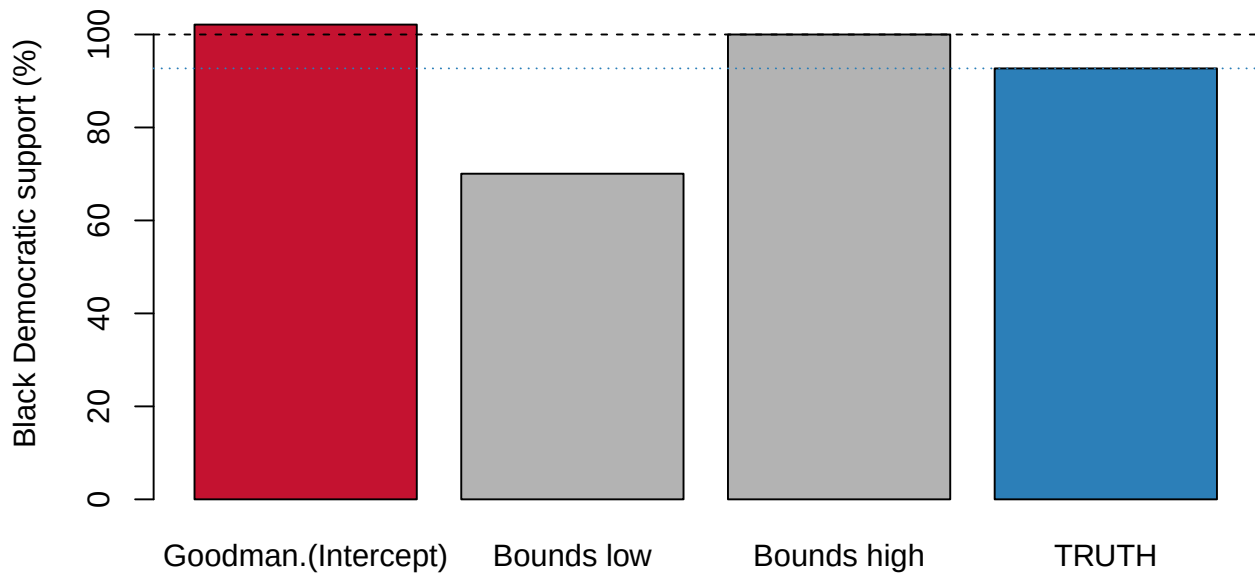
```
truth_black <- 100 * sum(p$black_dem) / sum(p$black)
truth_white <- 100 * sum(p$white_dem) / sum(p$white)

res <- data.frame(
  method = c("Homogeneous precincts", "Goodman regression",
             "Bounds (lower)", "Bounds (upper)", "TRUTH"),
  black = c(NA, est_black, 100 * max(lo), 100 * min(hi), truth_black),
  white = c(100 * sum(hw$dem) / sum(hw$voters), est_white, NA, NA, truth_white))

print(res, row.names = FALSE, digits = 3)
```

```
#>           method black white
#> Homogeneous precincts    NA 18.00
#> Goodman regression 102.1  3.30
#> Bounds (lower)    70.1    NA
#> Bounds (upper)  100.0    NA
#> TRUTH    92.7  7.14
```

```
barplot(c(Goodman = est_black, `Bounds low` = 100 * max(lo),
          `Bounds high` = 100 * min(hi), TRUTH = truth_black),
  col = c("#C41230", "grey70", "grey70", "#2c7fb8"),
  ylim = c(0, 110), ylab = "Black Democratic support (%)")
abline(h = 100, lty = 2); abline(h = truth_black, lty = 3, col = "#2c7fb8")
```



The truth is 92.7% for Black voters and 7.1% for white voters. This county is about as racially polarized as American politics gets, and every method in this lab said so. On the question these methods were built to answer — *is voting polarized here?* — they all got it right.

On the question *Callais* now asks — is it polarized by **race** rather than by **party**? — none of them took a position. They could not. What this file records is the party of the ballot each voter requested.

Now look at how they got there.

- **The bounds contained the truth**, as they must. They always do. That is their entire virtue and it cost them 29.9 points of precision.
- **Goodman missed by 9.4 points** and landed outside the range of possible numbers. Its white estimate, 3.3%, is less than half the true 7.1%.
- **Homogeneous precincts could not answer at all** for Black voters, and overstated white Democratic support by 10.9 points — 18.0% against 7.1%.

The uncomfortable part is that the method that was *most confident* was the one that was wrong, and its confidence was legible: R^2 of 0.982, a tight line, a clean plot. Nothing in the output warned anybody. **The impossibility was the only warning, and only because the truth happened to lie near a boundary.** In a county where Black Democratic support were 60%, Goodman could have missed by the same 9.4 points and returned a perfectly plausible 51%, and no one would ever have known.

Part 7 — Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Move the homogeneity threshold. Part 3 used 80%. Try 70%, or 60% (`hb <- p[p$pct_black >= 0.60, ]`). At some threshold you get a Black estimate. How close is it, and how much did you have to loosen the definition of “homogeneous” to get it? Would you defend that threshold under cross-examination?

Option B — Drop the biggest precinct. WELLSTON CENTER is the only precinct above 50% Black, so it is doing nearly all the work of anchoring the right-hand end of the line. Refit without it (`p2 <- p[p$precinct != "WELLSTON CENTER", ]`) and compare. What does the change tell you about how much of a fifty-year-old courtroom method rests on one polling place?

Option C — Predict, then check, precinct by precinct. The regression implies a Democratic share for each precinct (`fitted(fit)`). Compare those to what each precinct actually did (`p$pct_dem`), and then to what the truth columns say Black voters in that precinct actually did (`p$black_dem / p$black`). Where is the model worst, and is there a pattern to where it fails?

Option D — Turn it around. We validated the method where we could check it. Write down what would have to be true for today's result to transfer to the November general election, where nobody can check it. Then say how confident you would be, and why.

Option E — Your own question. 17 precincts, 16,024 voters, and a column called `black_dem` that real analysts never get.

What this lab cannot tell you

- **This would not establish *Gingles* preconditions two and three.** Before 29 April 2026 it would have. Since *Callais* a plaintiff has to show racial bloc voting that partisanship cannot explain, and the only outcome in this file is which party's ballot a voter asked for. We measured the confound. We did not break it.
- **A primary ballot is not a vote.** Requesting a Democratic ballot in May is not the same as voting for a Democrat in November, and primary electorates are smaller and more partisan than general ones — 18,133 people here against 82,114 in the general. We validated the *method*, on a real electorate, for a real choice. We did not measure how Houston County votes.
- **This is the easy case.** Polarization here is near-total, so every method reached the correct legal conclusion despite bad numbers. In a county at 55/45 the same errors would flip the answer. **Do not conclude that these methods are fine because they passed today.**
- **Two groups only.** 1,135 primary voters — Hispanic, Asian, American Indian, Other, Unknown — were dropped, because two-group ecological inference has nowhere to put them. They voted. The framework, not the data, excluded them.
- **Race on the form is still a measurement**, with the same one-box problem Weeks 12 and 14 raised.
- **We never ran King's method**, only the two it was designed to replace. It would do better here. It would also still be a model, and it would still be unfalsifiable in every county but this kind.

On the discussion board

By **Friday, 11:59 p.m.** No length requirement; a sentence or two is enough, so long as it shows you engaged with what we did. Any *one* of the usual four.

Four that would make good posts:

- A method returned 102% and had an R^2 of 0.982. What is the general lesson about trusting model fit?
- The bounds were guaranteed right and 30 points wide; the regression was precise and wrong. If you were the judge, which would you rather be handed, and what would you ask for instead?
- Every method got the *polarization* answer right and the *numbers* wrong. Does that make this a success story or a warning?
- *Callais* says a difference in party between racial groups “proves nothing.” Look again at the two truth numbers in Part 6 and ask what evidence could tell race and party apart here. If you cannot think of any, is the Court's requirement a demand for rigour or a demand that cannot be met?

Where the data came from

Houston County, Georgia, General Primary of 21 May 2024. Party of the primary ballot requested comes from the state's voter-history file; race is self-reported on the voter registration form; precinct is from registration extract 71754. All three are public records in Georgia. Working data for the 2026 Houston County redistricting matter.

Nothing individual-level is in the committed file — 16,024 voter records were collapsed to 17 precinct rows before anything was saved. **Rows whose race had been filled in by BISG rather than reported by the voter were dropped** before the collapse; grading one inference against another would have proved nothing. The build is in `data/build-data.R`.